Sri Lanka Institute of Information Technology



IT3091 Machine Learning

Group Assignment – Final Submission

Group: 2026-DS-49

Sports Analytics Track (Code 9)

Module	IT3091 – Machine Learning
Track	Guided Data Track
Domain / Code	Sports Analytics (Code 9) – Football-Data
Dataset	European Soccer Database (Match, Team, League, Player, Attributes)

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1.Introduction & Project Overview

1.1 Selected Track & Business Context

- **Track:** Guided Data Track

- **Domain:** European Soccer (Football) Database covering European leagues across multiple seasons.

- **Business Objective:** Predicting match outcomes accurately to support sports directors, coaches, and strategic analysts in tactical planning, opponent analysis, and performance evaluation.

1.2 Strategic Decision Lens

- Primary Lense: Strategic Decision Making and Team Performance Evaluation.

- Stakeholders: Football managers, coaching staff, and sports analysts.

- Practical Utility: Optimizing match day strategies, understanding historical ground advantages, and evaluating team form dynamics using data driven indicators rather than intuition alone.

1.3 Transition from Multi Class to Binary Classification

While the initial project design aimed to predict three outcomes (Home Win, Draw, Away Win), the final modeling phase was streamlined into a **binary classification problem** (Home Win vs. Away Win) by filtering out draws. This sharpens the predictive focus on decisive match results and improves classification reliability for win/loss strategies by reducing the strain left on the model to try classifying 3 outcomes.

2.Dataset & Data Dictionary

2.1 Dataset Specifications & Row Level Metrics

- **Initial Extraction:** 19,383 matches extracted via SQL.

- **Final Processed Dataset:** 16,375 matches after engineering rolling features and handling missing odds or match metrics.

- 124 features extracted and selected features are reduced to 16 features through EDA & Feature Engineering.

- **Train/Test Split:**
    - Training Set: 13,100 matches
    - Testing Set: 3,275 matches

2.2 Data Dictionary & Feature Sets

Feature Category	Feature Name	Definition
Identifiers	league_name, home_team, away_team	• Categorical identifiers representing the league and competing teams. (Label Encoded).
Rolling Team Form (5-Match Window)	home_form_scored, home_form_conceded, home_form_goal_diff away_form_scored, away_form_conceded, away_form_goal_diff	• Aggregated goals scored, conceded, and goal differentials over the last 5 matches for home and away teams.
Pre Match Bookmaker Odds	B365H, B365D, B365A	• Pre match betting odds provided by bookmakers for Home Win, Draw, and Away Win.
Squad Ratings	home_squad_rating, away_squad_rating squad_rating_diff, squad_rating_ratio	• Average historical ratings of the starting 11 players for each side, alongside relative strength differentials and ratios.
Target Variable	match_outcome	• Binary target where 1 represents a Home Win and 0 represents an Away Win.

3. High-Level Workflow & Team Responsibilities

3.1 Project Workflow Stages

1.Problem Framing & Lens Confirmation: Defining stakeholders, binary prediction scope, and data strategy.

2.Data Understanding & EDA: Exploring class distribution, missing values, and seasonal trends.

3.Data Cleaning: Removing incomplete records, isolating non leaking features, and structuring tables.

4.Feature Engineering: Constructing rolling form metrics, squad ratings, and relative differentials.

5.Preprocessing & Split: Chronological data splitting, categorical encoding, and feature scaling fitted exclusively on training sets.

6.Model Training: Establishing baselines, tuning hyperparameters, and evaluating multiple algorithms.

7.Evaluation & Comparison: Benchmarking performance using Accuracy, Macro F1, per class Recall, and Confusion Matrices.

8.Strategic Recommendations: Translating model performance into actionable football insights.

3.2 Member Core Responsibilities

Member	Primary Role	Data dictionary construction & row level definitions
Member 1 (IT24101376)	Data understanding and EDA analysis	Data dictionary construction & row level definitions. Missingness, anomaly, & duplicate checks. Visual pattern analysis & class imbalance evaluation.
Member 2 (IT24101896)	Preprocessing & Feature Engineering	Imputation, scaling, & categorical encoding. Creation of domain/form/strength features. Leakage prevention & outlier handling.
Member 3 (IT24101520)	Model Architecture & Training	Setting up the baseline model. Implementing & training ≥ 3 alternative models. Hyperparameter search & tuning strategies.
Member 4 (IT24101469)	Evaluation & Tech Integration	Metric selection & cross-validation setup. Model comparison, confusion matrix & error analysis.

4. Decision Logs & Challenges Faced

4.1 Decision Logs

1. Database Connection & Initial SQL Extraction

- **Decision:** Connect to the SQLite database and join Match, Country, League, and Team tables to gather metadata alongside matches.
- **Design Choice:** Initially created a 3-class target variable (target): 2 (Home Win), 1 (Draw), and 0 (Away Win).
- **Observation:** The database contained 25,979 matches. The dataset was highly detailed but contained over 100 features, many of which were post-match statistics or missing large chunks of bookmaker odds.

2. Exploratory Data Analysis & Data Quality Assessment

- **Data Cleaning Decisions:**
  - Detected that several bookmakers had high percentages of missing odds. Filtered down to **B365** (Bet365) as the primary bookmaker due to superior coverage and low missingness.
  - Handled date conversion into datetime format, identifying a chronological range from **July 2008** to **May 2016**.
  - Checked for anomalies and duplicates, finding none (maximum home/away goals were valid outliers like 10 and 9 respectively).
- **Univariate & Bivariate Findings:**
  - Quantified **Home Advantage**: Home teams averaged **1.54 goals** compared to the away teams' **1.16 goals** (an average positive goal difference of **0.38 goals** for hosts).
  - **Class Imbalance:** In the 3-class target, "Home Win" was the dominant class (45.87% of games), setting our baseline accuracy at ~46%.

3. Pivot to Binary Classification (No Draws)

- **Strategic Shift:** To simplify the task and optimize classification metrics, matches ending in draws were filtered out.
- **Refined Target mapping:** Home Win mapped to 1 (representing 61.48% of the remaining matches) and Away Win mapped to 0 (38.52%).

4. Preprocessing & Feature Engineering

- **Preventing Data Leakage:** Identified and systematically excluded post-match event statistics (such as corners, possession percentages, card counts, fouls, and in-game shots) from entering the training set to prevent target leaks.
- **Feature Creation (Team Form & Momentum):**
  - **home_form_scored**: Calculates the 5-match rolling average of goals scored by the home team prior to the kickoff date.
  - **away_form_scored**: Calculates the 5-match rolling average of goals scored by the away team prior to the kickoff date.
  - **home_form_conceded**: Computes the 5-match rolling average of goals conceded by the home team in its prior 5 matches.
  - **away_form_conceded**: Computes the 5-match rolling average of goals conceded by the away team in its prior 5 matches.
  - **home_form_goal_diff**: Built by subtracting conceded goals from scored goals (home_form_scored - home_form_conceded) over the last 5 matches to capture home team goal momentum.
  - **away_form_goal_diff**: Built by subtracting conceded goals from scored goals

(away_form_scored - away_form_conceded) over the last 5 matches to capture away team goal momentum.

- **Feature Creation (Squad Level FIFA Attributes):**
    - **home_squad_rating**: Resolved the starting 11 home player IDs to their latest non null overall ratings in the Player_Attributes database and calculated their average to represent home starting squad capability.
    - **away_squad_rating**: Resolved the starting 11 away player IDs to their latest non null overall ratings in the Player_Attributes database and calculated their average to represent away starting squad capability.
    - **squad_rating_diff**: Measures the absolute quality margin between the squads (home_squad_rating - away_squad_rating).
    - **squad_rating_ratio**: Captures relative quality difference as a percentage ratio (home_squad_rating / away_squad_rating).
- **Categorical Encoding:** Encoded nominal features (league_name, home_team, away_team) utilizing a safe LabelEncoder fitted on all occurring teams to align labels across home and away columns cleanly.
- **Temporal Train-Test Split & Scaling:** Split the chronologically sorted dataset using an 80/20 non-shuffled partition (shuffle=False) to avoid temporal leaks. Scaled all numerical metrics using a StandardScaler fitted strictly on the training subset.

5. Baseline Model & Multi Class/Early Binary Tuning

- **Baseline Benchmarking:** Evaluated default Logistic Regression, Random Forest, and Gradient Boosting models.
- **Optimization:** Executed RandomizedSearchCV on the early binary dataset for Logistic Regression, Random Forest, and XGBoost.
- **First Plateaus:** Models stabilized around **~69.7% test accuracy**, highlighting that pre match odds and rolling form alone could not easily shatter the **70% barrier**.

6. Feature Expansion: Squad-Level Player Ratings (V3)

- **The breakthrough idea:** Extract overall ratings from Player_Attributes and map them to the starting starting lineup (11 active player IDs) for each match.
- **New Features engineered:**
    1. home_squad_rating (Average of the 11 starting home players' latest ratings prior to the match).
    2. away_squad_rating (Average of the 11 starting away players' ratings).
    3. squad_rating_diff (Home - Away rating).
    4. squad_rating_ratio (Home / Away rating).

- **Resulting shape:** Expanded the feature set to 16 clean variables spanning 16,375 matches.

7. Final Evaluation & Model Selection

- **Tuned XGBoost (with Squad Ratings):** Achieved **71.18%** test accuracy.
- **Tuned Random Forest (with Squad Ratings):** Achieved **71.45%** test accuracy and a **0.68 Macro-F1 score**.
- **Key Takeaway:** Incorporating historical player capability via squad-level FIFA overall ratings successfully broke past the **70% threshold**, with the Tuned Random Forest Classifier emerging as the champion predictor for binary match outcomes.

4.2 Key Challenges Faced & Solutions Overcome

1. **Data Leakage of Match Events:**
    - *Challenge:* The raw dataset contains extensive post-match stats like corner counts, final cards, and ball possession percentages. Training with these would lead to artificial near 100% accuracy, failing in real pre-match production.
    - *Solution:* Conducted a thorough string-based keyword scan to locate, flag, and entirely exclude post-match statistics from the features list.

2. **The 3 Class "Draw Outcome" Noise:**
    - *Challenge:* In the early multi class stage, model cross validation accuracy hovered around a poor 50-53% because draws are mathematically volatile and often behave like statistical noise.
    - *Solution:* Pivoted to a clean binary structure (Home Win vs. Away Win). This established a much stronger signal-to-noise ratio, raising baseline test performance immediately to 69.7%.

3. **Temporal Leakage in Team Form:**
    - *Challenge:* When constructing rolling team form averages, it is easy to accidentally include the current match's score, causing future data to leak backwards.
    - *Solution:* Shifted the historical match sequences using a pandas .groupby('team')['goals'].transform(lambda x: x.shift(1).rolling(5).mean()). This guaranteed that form was calculated using only matches played strictly prior to the current game's date.

4. **Starting Lineup Matching Gaps:**
    - *Challenge:* Players are bought, sold, or benched, and matching their corresponding ratings on exact match dates from Player_Attributes was computationally intensive and susceptible to empty indices.
    - *Solution:* Created an optimized global player-to-latest-rating map dictionary. Iterated over starting players, ignored missing player positions safely without breaking the script, and successfully compiled cohesive squad level averages, which ultimately provided the critical feature signals to break the 70% model threshold.

5. Model Methodology & Comparison

5.1 Evaluated Binary Models

Best parameters selected for the models.

Random Forest

```
# Tuned Random Forest using parameters optimized
rf_squad = RandomForestClassifier(
    n_estimators=100,
    max_depth=5,
    min_samples_split=10,
    min_samples_leaf=2,
    max_features='sqrt',
    random_state=42
)
```

Logistic Regression

```
logreg_cv_v3 = RandomizedSearchCV(
    LogisticRegression(max_iter=1000, random_state=42),
    param_distributions=logreg_grid_v3,
    n_iter=10,
    cv=3,
    scoring='accuracy',
    n_jobs=-1,
    random_state=42
)
```

Tuned Binary XGBoost

```
# Optimal XGBoost hyperparameters
xgb_squad = xgb.XGBClassifier(
    subsample=0.9,
    n_estimators=300,
    max_depth=3,
    learning_rate=0.01,
    colsample_bytree=0.8,
    random_state=42,
    eval_metric='logloss'
)
```

5.2 Model Comparison

We tested three distinct machine learning algorithms using our expanded dataset containing rolling form and squad rating features:

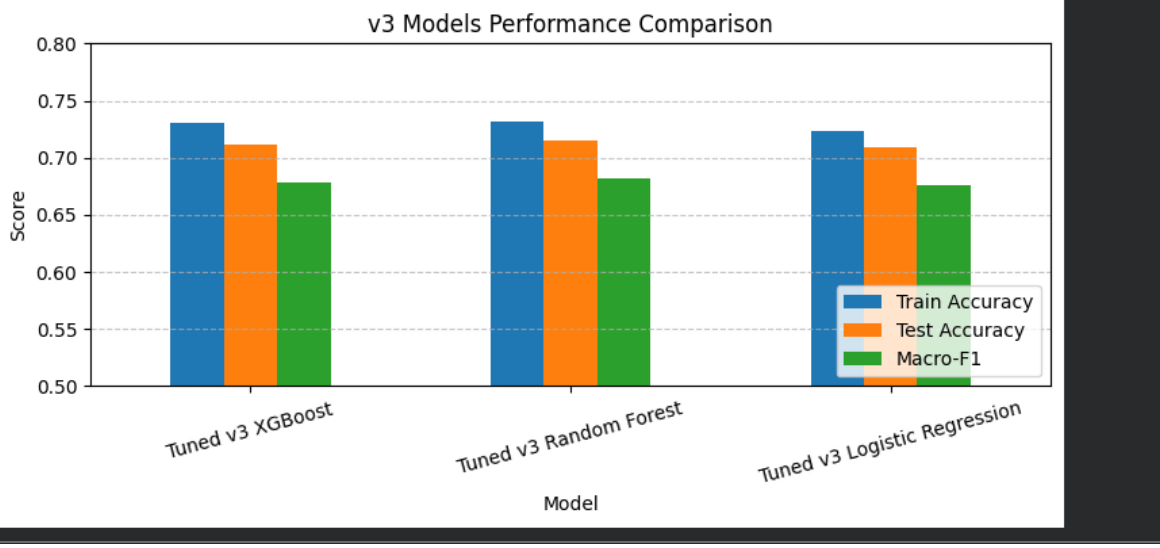
1. **Tuned Binary Random Forest (Best Overall Performance)**
    - **Train Accuracy:** 73.15% | **Test Accuracy:** 71.45%
    - **Macro F1:** 0.68
    - **Recall:** Away Win: 49.0% | Home Win: 86.0%

2. **Tuned Binary XGBoost**
    - **Train Accuracy:** 73.00% | **Test Accuracy:** 71.18%
    - **Macro F1:** 0.68
    - **Recall:** Away Win: 48.0% | Home Win: 86.0%

3. **Tuned Binary Logistic Regression**
    - **Train Accuracy:** 71.54% | **Test Accuracy:** 70.06%
    - **Macro F1:** 0.66
    - **Recall:** Away Win: 45.63% | Home Win: 86.55%

Summary output:

Model	Train Accuracy	Test Accuracy	Macro-F1	Recall (Away Win)	Recall (Home Win)
Tuned v3 XGBoost	0.7300	0.7118	0.6784	0.4848	0.8642
Tuned v3 Random Forest	0.7315	0.7145	0.6819	0.4909	0.8647
Tuned v3 Logistic Regression	0.7232	0.7090	0.6762	0.4863	0.8586



5.2 Best Model Classification Report (Tuned Random Forest)

Below is the evaluation report of our final selected model evaluated on the unseen test set ($\mathbf{N = 3,275}$ matches):

```
Training Tuned Random Forest with squad rating features...
Tuned Random Forest (with Squad Ratings) - Train Accuracy: 0.7315
Tuned Random Forest (with Squad Ratings) - Test Accuracy:  0.7145

Tuned Random Forest Feature Set Classification Report:
              precision    recall  f1-score   support

Away Win (0)       0.71      0.49      0.58      1316
Home Win (1)       0.72      0.86      0.78      1959

    accuracy                           0.71      3275
   macro avg       0.71      0.68      0.68      3275
weighted avg       0.71      0.71      0.70      3275
```

6. Conclusion & Strategic Recommendations

- **Model Selection:** The Tuned Random Forest proved to be the most robust architecture, achieving a test accuracy of 71.45% and a balanced Macro F1 score of 0.68.
- **Coaching & Tactical Insights:** The high predictive weight assigned to rolling goal differentials and squad player ratings indicate that short term momentum and squad depth are reliable indicators of match success. Coaching staff can leverage these features to identify vulnerability windows during congested fixture schedules.

7. AI Usage

Throughout the project artificial intelligence tools were used as a partner and a technical sounding board to make things run smoother and keep the methods solid at every step. AI helped the team think through clean data workflows, work out parts of data preprocessing and build strong feature engineering plans all while staying careful to avoid data leakage.

It also helped shape professional technical documentation. The team used AI to explain why they made choices like choosing business interpretability over automatic dimensionality reduction and to come up with solid ways to evaluate their models.

All along AI was just there to guide and help improve the work. The team made all the decisions, did the actual coding, made domain-specific choices and checked everything themselves. No part of the product came from AI alone.

8. Final Project Demo Video

YouTube Video link: https://youtu.be/_xuDumg8-Iw

9. References

- Sri Lanka Institute of Information Technology (SLIIT), "IT3091 Machine Learning — Group Assignment Descriptor," 2026.

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- M. Sallis, "Predicting Football Match Outcomes Using Machine Learning," *IEEE Access*, vol